CRYPTOGRAPHY LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
Instructor:	Ray Lampano, Jnr.
Section Number:	15480/15481
Student Name:	Donaven Bruce
Student ID:	[Redacted]
Submission Platform:	Canvas
Report Date:	2026-02-21
Report Version:	1.0
Lab Title:	Cryptography
Module Reference:	Cryptography
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-02-21
Time Spent:	Approximately 1 hour

ENVIRONMENT

Platforms and Systems:

- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox.
- Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Controller), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Domain Member), and PLABKALI01 (Kali 2019.2 - Linux Kali Workstation).
- The work centered mainly on PLABDM01 using File Explorer, Internet Explorer, Crypt4Free, HashCalc, MD5 Calculator, and GiliSoft Full Disk Encryption.
- Remote access to the virtual machines was provided through the course platform.

Context and Constraints:

- The module focused on file/folder encryption (Crypt4Free), hash calculation (HashCalc and MD5 Calculator), and full disk encryption (GiliSoft).
- Practice Labs flagged the module as legacy content using deprecated software, so behavior could vary and some steps were best interpreted as guided sandbox demonstrations.
- Windows Updates were disabled in the Practice Labs environment, so software behavior reflected a controlled legacy setup rather than a current hardened enterprise baseline.
- The module aligned most closely with exam objectives 3.1 (Information Security Controls).

Authorization and Scope:

- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes.
- No public IP space, real organizations, or unauthorized targets were involved.
- No production infrastructure, third-party systems, or live services were contacted.
- The actions documented here were performed for defensive understanding, cryptography tool usage, and ethical hacking education within the course environment.
- Restricted material, live credentials, and personal data are omitted or redacted from this report.

Work Objectives:

- Prepare files and folders for encryption.
- Install and use Crypt4Free to encrypt/decrypt files and folders with password protection.
- Download, install, and use HashCalc to compute multiple hash algorithms.
- Install and use MD5 Calculator to verify file integrity by comparing hashes.
- Perform full disk encryption using GiliSoft Full Disk Encryption on the C: drive.

Cryptography Concepts and Defensive Framing:

- The lab identified three core phases of cryptography: file/folder encryption, hash verification, and full disk encryption.
- Tools demonstrated both commercial (Crypt4Free, GiliSoft) and utility-based (HashCalc, MD5 Calculator) approaches, showing how encryption and hashing protect data at rest.
- Prevention concepts reviewed included strong passwords, full disk encryption, and hash verification to ensure file integrity, reinforcing that cryptography defense requires both encryption and integrity checking.

TECHNICAL ANALYSIS
File and Folder Encryption with Crypt4Free on PLABDM01:
– On PLABDM01, File Explorer was used to create the C:\labfiles folder. – Internet Explorer was used to download md5calculator_setup.exe and crypt4free setup.exe from the Intranet Hacking Tools page. – Crypt4Free (Advanced Encryption Package) was installed and launched. – The C:\labfiles folder was selected and encrypted using the password Passw0rd (Encrypt Now!). – Each file received a .aep encrypted counterpart with a key icon. – Decryption was tested by double-clicking md5calculator_setup.exe.aep, entering Passw0rd, and successfully restoring the original file.
Hash Calculation and Verification with HashCalc and MD5 Calculator:
– HashCalc was downloaded as hashcalc.zip, extracted, and installed. – The md5calculator_setup.exe file was loaded into HashCalc and multiple hashes (MD5, SHA1, SHA256, etc.) were calculated. – MD5 Calculator was downloaded as md5calculator_setup.exe and installed. – The same file was added to MD5 Calculator and the MD5 hash was computed. – The MD5 hash from HashCalc was copied and pasted into MD5 Calculator's Verify field. – The Compare button confirmed both MD5 values matched ("OK, two MD5 values are equal").
Full Disk Encryption with GiliSoft on PLABDM01:
– GiliSoft Full Disk Encryption was downloaded as full-disk-encryption.exe and installed. – The application was launched and the C: [System] drive was selected for encryption. – The password Passw0rd was set and the full disk encryption process was started.

KEY COMMANDS, TOOLS, AND ARTIFACTS BY SYSTEM
PLABDM01 — Cryptography Tools:
Crypt4Free (Advanced Encryption Package) - Encrypt / Decrypt with password Passw0rd HashCalc - Load file → Calculate (MD5, SHA1, SHA256, etc.) MD5 Calculator - Add Files → Calculate → Verify MD5 (compare with HashCalc) GiliSoft Full Disk Encryption - Select C: drive → Encrypt with password Passw0rd
Primary Tools and Artifacts Observed:
– Crypt4Free successfully encrypted/decrypted files in C:\labfiles (created .aep files with key icon). – HashCalc computed multiple hashes for md5calculator_setup.exe (MD5, SHA1, SHA256, etc.). – MD5 Calculator verified the MD5 hash matched the value from HashCalc ("two MD5 values are equal"). – GiliSoft Full Disk Encryption started encrypting the C: system drive with password Passw0rd. – The most important evidence of successful cryptography operations was the encrypted .aep files, matching MD5 hashes between tools, and the initiated full disk encryption process.

DEFENSIVE MAPPING AND SUMMARY
Technique Summary:
– The completed work covered file/folder encryption (Crypt4Free), hash calculation and verification (HashCalc and MD5 Calculator), and full disk encryption (GiliSoft). – The recurring theme across the module was that cryptography protects data at rest through encryption and ensures integrity through hashing.
MITRE ATT&CK Mapping:
– Defense Evasion: T1027 Obfuscated Files or Information - file encryption with Crypt4Free. – Impact: T1486 Data Encrypted for Impact - full disk encryption with GiliSoft. – Collection: T1119 Automated Collection - hash verification with HashCalc/MD5 Calculator. – Credential Access: T1555 Credentials from Password Stores - protection of sensitive data at rest.
Detection Opportunities:
– Unexpected encrypted files (.aep), full disk encryption processes, or unusual hashing activity can indicate data protection or ransomware activity. – Endpoint monitoring for Crypt4Free, GiliSoft, or hash tool execution provides visibility. – File integrity monitoring tools can detect unauthorized encryption or hash changes.
Defensive Controls:
– Use strong encryption tools (Crypt4Free/GiliSoft) with complex passwords for sensitive files and drives. – Implement full disk encryption (BitLocker or third-party tools) on all endpoints. – Verify file integrity with hashing tools (HashCalc/MD5 Calculator) before installation. – Enforce least-privilege, regular patching, and endpoint detection/response (EDR) for cryptography tools.
Completion Verification:
– Work was completed across file/folder encryption, hash calculation, MD5 verification, and full disk encryption. – Completion was evidenced by successful encryption/decryption of labfiles, matching MD5 hashes, and initiated full disk encryption on C: drive. – The actual tools, steps, and outputs used in the lab are preserved directly in this report for accurate documentation and reproducibility.
Key Findings:
1) File and folder encryption (Crypt4Free) effectively protects sensitive data at rest. 2) Hash tools (HashCalc and MD5 Calculator) allow reliable verification of file integrity. 3) Full disk encryption (GiliSoft) secures entire drives, including the system partition. 4) Strong passwords and proper tool usage are essential for effective cryptography. 5) The strongest defensive takeaway was that combining encryption, hashing, and full disk encryption provides robust protection for data at rest.
Issues / Deviations / Notes:
– The lab used deprecated software and a legacy environment, so exact tool behavior could vary. – Full disk encryption process was started but takes several minutes to complete. – All activities were performed only inside the isolated Practice Labs sandbox.
Reflection:
– This lab tied together file encryption, hash verification, and full disk encryption into a complete cryptography workflow for protecting data at rest. – The strongest takeaway was that cryptography is dangerous to ignore because unencrypted data at rest is easily accessible to attackers, making encryption and integrity checking essential for defense.